GDG Windsor – Build with AI

Smart Auomation, Robotics, and Vision

DataRealm Inc.

Hackathon.HumanoidAppLab.com

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About Me

- Automation Specialist on the Humanoid Application Lab team at DataRealm
- PhD student at the University of Windsor
- Focus on simulations, digital twins, and robotics
- Will be available for help throughout the hackathon
  - Questions
  - Ideas
  - Clarification
  - Guidance
- SRice@DataRealmInc.com



DataRealm

- Founded in Windsor in 2001
- Controls Engineering
  - PLC/HMI
  - Electrical design
- Information Systems
  - Manage plant floor and control systems
- Custom Software

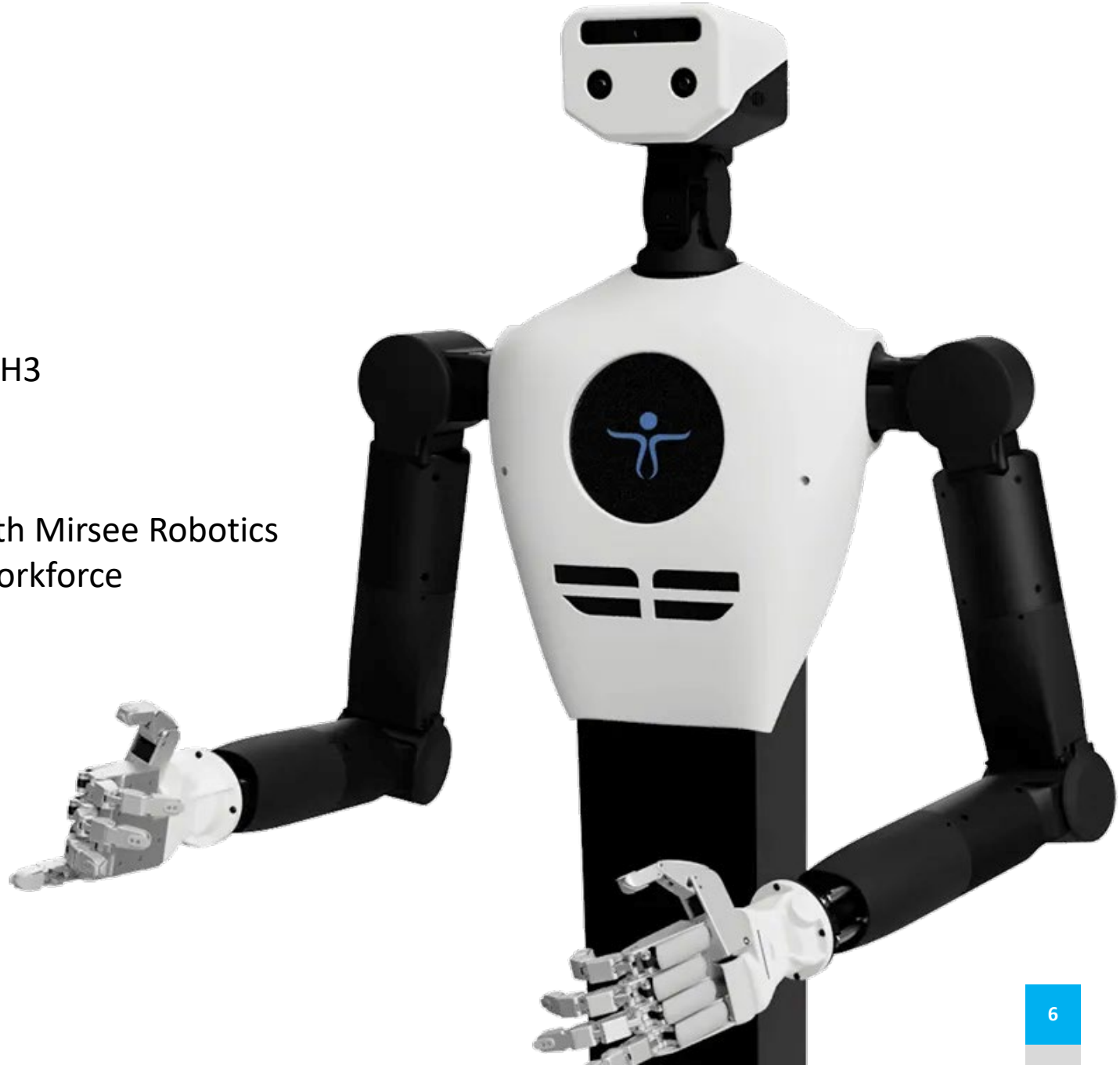


ABB, FCA (Fiat Chrysler Automobiles), Ford, G&W (Engineered to order. Built to last.), GM, GROB, Hiram Walker, Honda, Jamieson (Natural Sources), Kautex (A Textron Company), Linamar (Power to Perform), Onbelay Automotive Inc, REKO International Group Inc., Safran, sciemetric, thyssenkrupp

- Division of DataRealm started this year
- Working to bridge humanoids from research labs and into the workplace
- Variety of backgrounds
  - Computer Science
  - Mechanical Engineering
  - Electrical Engineering



- Developing humanoid robots in Cambridge
  - Most components produced in Canada
- On their third generation of humanoid, the MH3
  - Developing the MH4
- Humanoid Application Lab working closely with Mirsee Robotics to bring Canadian robots into the Canadian workforce



Technologies Driving Humanoids

- Reinforcement Learning (RL), Vision-Language Action Models (VLAs), Vision-Language Models (VLMs)
  - RL – Time consuming to train on specific tasks; black-box
  - VLAs – Time consuming to train on more complex tasks; black-box
  - VLMs – Slower inference but general; lack fine-grain control; more transparent
- Auxiliary features like dedicated vision models, motion planning, etc.
  - YOLO vision models
  - Inverse kinematics for manipulation
  - Navigation with lidar, SLAM, and depth-camera

Humanoids in Industry

- Practically non-existent
  - Some companies deployed in limited capacities and in limited roles
- Why is this?
  - Most humanoids still in a “research phase”
  - Software compatibility
  - No standards
  - Highly unpredictable
- Industry needs:
  - Standards, repeatability, and trust
  - Heuristic control programs like behaviour trees or ladder logic

Challenges Integrating Humanoids

- How can we bridge the gap between the control technologies used for humanoids and the needs for industry?
- How can we ensure repeatability for humanoids?
  - Cycle times?
  - Safety constraints?
  - Fault detection and recovery?
- How can we interface with PLCs and other existing infrastructure?
- How can workers debug humanoids?
- How can different types of humanoids work with each other?
- How can different humanoids be easily programmed?

Hackathon Template Project

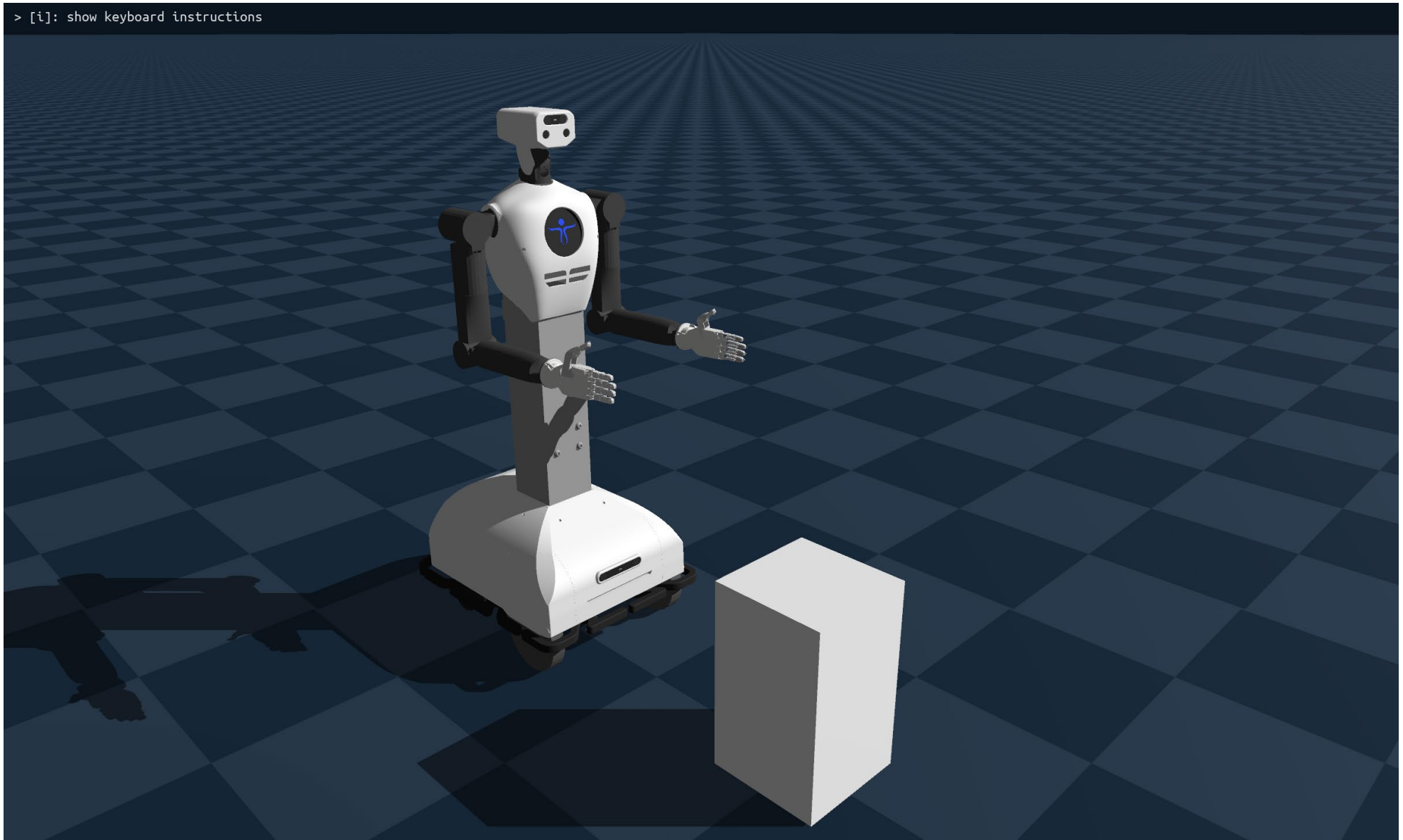
- Provides a Python simulator with the MH3
  - Built on Genesis World
  - No external robot dependencies
  - Inverse kinematics solving
- Provides a testing web dashboard
- Works on Windows, Mac, and Linux



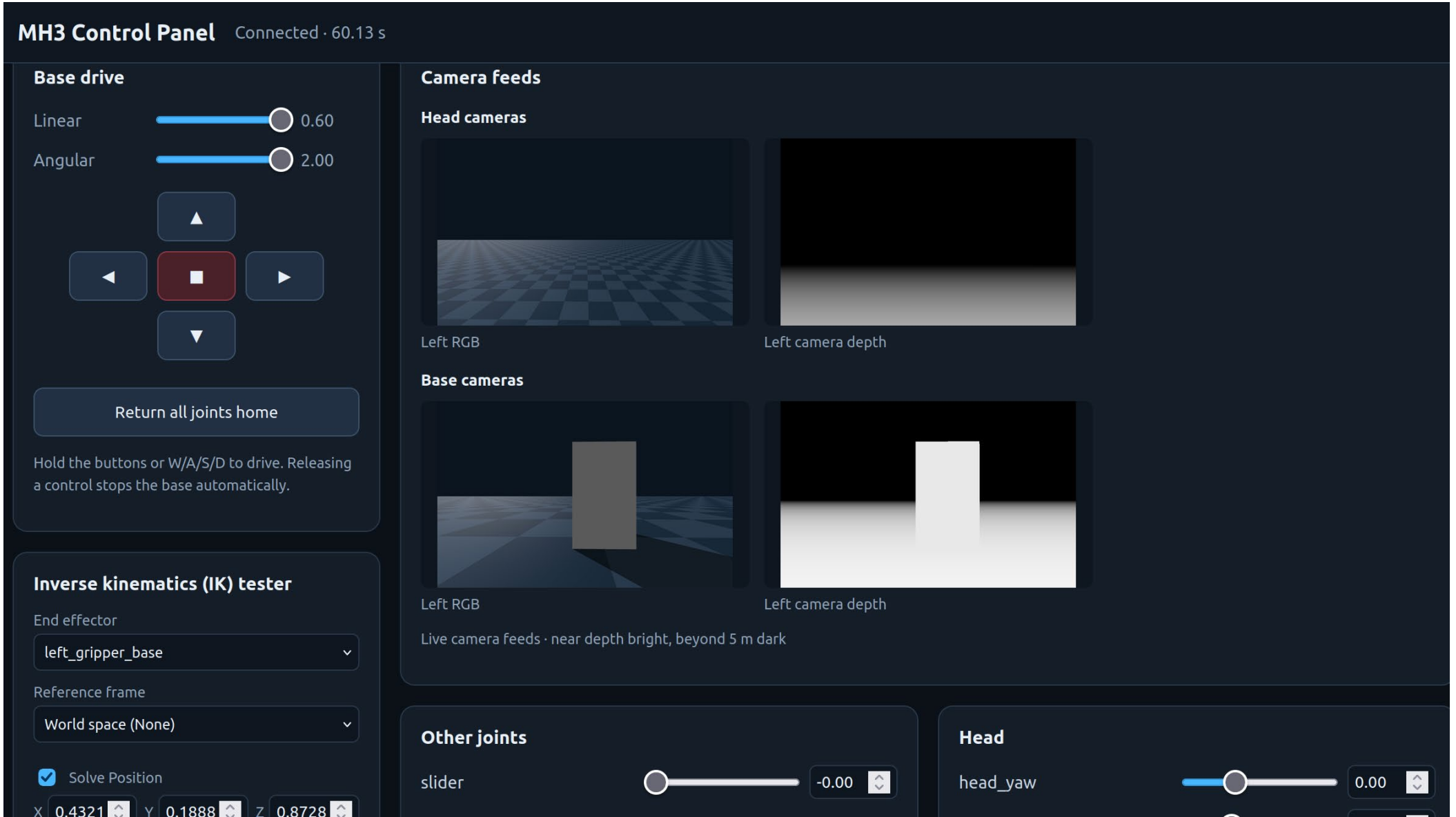
Genesis

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Hackathon Template Project



Hackathon Template Project



Objectives

- Emboddied AI is the next frontier of robotics
  - Every major AI company getting into physical AI
- Humanoid robots provide the most posibilities for emboddied AI
  - Currently difficult to work with and unpredictable
- **Close the gap between emboddied AI and industrial applications**
  - Robots working with humans
  - Robots working with other robots
  - Robots working with other technology
  - Balancing understandability, repeatability, accuracy, and intelligence

FAQ

- Do I have to use the template?
  - Nope! You are free to approach the hackathon challenges in any way you want
- Does my submission need to revolve around humanoid robots?
  - Nope! Industrial settings have many forms of robotics and automation technology, from industrial arms, vision systems, PLCs, robot dogs, humanoids, and more. You are not bound to one form of technology!
- If I work with a humanoid, does it have to be the MH3?
  - Nope! Mirsee Robotics kindly provided the model of the MH3 for this hackathon, but you can work with any humanoid robot you wish!
- If I use the template, can I use a different robot than the MH3 with it?
  - It would require more work on your end, but yes, you can rework the template to use anything you want!
- What is the license of the template project?
  - The code is MIT licensed; the MH3's model is free to use for anything related to this hackathon

FAQ

- Some of the topics seem they aren't directly related to "emboddied AI"?
  - Working with emboddied AI has a lot of aspects.
  - We need to develop simulations and digital twins, ensure proper training both for machines and humans, have strong ability to communicate between machines and humans, and be able to troubleshoot and debug when things go wrong.
  - Using AI tools in these ways lays all the necessary groundwork towards one day bringing emboddied AI into industry, so while they may not directly be "emboddied AI" themselves, they are an equally important part of it
- Will I be judged better or worse based on what topic I choose?
  - No! All listed topics were chosen as they are highly important, and we are excitied to see how you approach each and every one of them!
- Can I choose a topic of my own!
  - Yes! Anything that fits the theme of building solutions that would impact the industries of Windsor-Essex is fair game! If you have any questions about your ideas, don't hestitate to reach out!

Wrap-Up

- Be creative and have fun!
  - Learn new things about robotics and industry!
  - Help shape how we bring advanced AI into industry!
- Myself and DataRealm's Humanoid Application Lab will be at DevFest
  - Will be speaking on some of our own internal solutions to some of these challenges

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